EDGE AI & ROBOTICS · CONVERSATIONAL AI · COURSE PROJECT

CogniWatt

Learning the economics of cognition for edge robots — teaching a robot when to compute, when to ask, and when to just act.

SUBMITTED BY

NAME	ROLL NUMBER	WORKSTREAM
Adit Uppal	102373004	R2 — dialogue & measurement
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PROJECT LINKS

REPOSITORY	https://github.com/jayden-crypto/cogniwatt
DATASET	https://github.com/jayden-crypto/cogniwatt/tree/main/data

REPORT CONTENTS

PART I · PP. 2–19

CogniWatt

Headline result, training stability, entropy annealing, ablations, cloud-price derivation, hardware grounding, and four silent failures.

PART II · PP. 20–32

The Saturation Ceiling

28,920 measured responses over 660 scenarios and 5 model configurations, with a three-annotator human audit of the ground truth.

Simulation only · no physical hardware

Submitted 10 November 2026

EDGE AI & ROBOTICS · WORKSTREAM R1 · RL & ENVIRONMENT

CogniWatt

Teaching a robot the economics of its own thinking — when to compute, when to ask, and when to just act.

Owner Jayden (R1) **Status** complete — all channel constants measured, scenario set human-audited

Code 10,900 lines Python · 13 + 18 tests **Compute** 176 runs · ≈20.0 h on M2 Air

Team 2 people, 2 workstreams

Headline result

A single RL policy decides not just *what to do* but *how much compute to spend deciding*. Measured against real quantised models on a human-audited scenario set, the answer turns out to be **almost none** — and that inversion is the result.

0.968	**0.507**
success, cooperative human	success, offline and alone
±0.001 over 6 seeds · heuristic 0.964	±0.013 over 6 seeds · heuristic 0.316

Both rows are trained and evaluated on R2's audited artefact (`audited_321_contested_dropped`): every evidence tier measured on the same 321 scenarios, each one resolved identically by three independent human annotators. Reported as mean ± se **over seeds** — every one of the 6 seeds converged, so this is the whole population of runs, not a survivor-selected subset.

CONDITION	AGENT	SUCCESS	RETURN	ENERGY (J)	QUERIES / EPISODE
cooperative human	**ppo (6 seeds)**	**0.9680 ± 0.0005**	**+0.295 ± 0.015**	**162.3**	**1.06**
cooperative human	greedy_value	0.9640	-0.439	163.8	2.48
offline + alone	**ppo (6 seeds)**	**0.5070 ± 0.0129**	**-7.529 ± 0.221**	**165.2**	**2.48**
offline + alone	greedy_value	0.3160	-24.194	314.3	17.88

Compute cannot buy its way out of referential ambiguity

No model a domestic robot can reach — 0.5B, 3B or 7B, at Q4 or Q8, given unlimited queries or unlimited reasoning tokens — resolves these referring expressions confidently enough to act on. Asking the person does. So with a cooperative human available the decision problem is **degenerate**: one clarifying question reaches confidence, no model tier can, every agent converges on "ask once" and ties at 0.968, and on-device inference falls to **1 %** of evidence purchased.

That is not a failure of the environment — it is the measurement this project was built to make, and it inverts the premise it started from. **Dialogue is not the fallback for when compute runs out; compute is the fallback for when dialogue is unavailable.**

Which is exactly where the learned policy earns its place. Offline and alone, no single action suffices: the agent must sequence cheap, weak evidence under a budget, and it beats the strong heuristic by **19.1 points of success** and the best baseline by **+1.75 return**.

Full comparison — training devices

> Devices: `m2_air, orin_nano, orin_nx, pi5`

AGENT	SUCCESS	RETURN ± SE	ENERGY (J)	LATENCY (S)	ASKS	TINY / BIG / CLOUD
oracle	**1.000**	+3.34 ± 0.06	154.9	16.7	0.00	0.00 / 0.00 / 0.00
ppo[r2final321_s0]	**0.955**	**+0.17 ± 0.15**	**159.1**	**17.7**	**1.00**	**0.07 / 0.00 / 0.01**
greedy_value	**0.960**	-0.33 ± 0.15	159.6	17.9	0.88	0.05 / 0.00 / 1.57
always_ask	**0.957**	+0.24 ± 0.14	158.3	17.6	1.00	0.00 / 0.00 / 0.00
always_offload	**0.749**	-2.82 ± 0.28	157.6	17.2	0.00	0.00 / 0.00 / 2.00
always_big	**0.439**	-14.88 ± 0.42	316.8	34.5	0.00	0.00 / 2.69 / 0.00
always_tiny	**0.433**	-10.42 ± 0.33	208.6	23.2	0.00	4.00 / 0.00 / 0.00
no_think	**0.376**	-9.19 ± 0.31	156.1	16.8	0.00	0.00 / 0.00 / 0.00
random_valid	**0.041**	-30.32 ± 0.18	430.2	44.2	1.17	1.29 / 0.94 / 1.40
random	**0.028**	-31.97 ± 0.18	430.0	45.5	1.27	1.37 / 1.40 / 1.35

Zero-shot transfer to unseen hardware

> Devices: `jetson_nano_2gb, rtx3060` — never seen in training. The policy is conditioned only on datasheet features (TOPS, bandwidth, TDP, battery, NPU flag, RAM), never on the cost table.

AGENT	SUCCESS	RETURN ± SE	ENERGY (J)	LATENCY (S)	ASKS	TINY / BIG / CLOUD
oracle	**1.000**	+3.34 ± 0.06	154.9	16.7	0.00	0.00 / 0.00 / 0.00
ppo[r2final321_s0]	**0.958**	**+0.14 ± 0.14**	**162.9**	**17.6**	**1.00**	**0.01 / 0.00 / 0.01**
greedy_value	**0.960**	-0.48 ± 0.15	165.6	17.9	0.88	0.02 / 0.00 / 1.57
always_ask	**0.957**	+0.13 ± 0.14	163.0	17.6	1.00	0.00 / 0.00 / 0.00
always_offload	**0.749**	-2.87 ± 0.28	159.4	17.2	0.00	0.00 / 0.00 / 2.00
always_big	**0.472**	-9.22 ± 0.32	229.2	17.6	0.00	0.00 / 1.50 / 0.00
always_tiny	**0.426**	-10.07 ± 0.32	204.3	20.9	0.00	3.98 / 0.00 / 0.00
no_think	**0.376**	-9.19 ± 0.31	156.1	16.8	0.00	0.00 / 0.00 / 0.00
random_valid	**0.022**	-27.82 ± 0.16	383.5	36.2	1.02	1.06 / 0.54 / 1.17
random	**0.025**	-29.10 ± 0.18	376.7	34.7	1.32	1.38 / 0.74 / 1.35

Training stability — a solved limitation

This report previously carried an open limitation: *"6 of 10 non-ablation runs converge; 4 collapse to ~0.34 … raising the entropy bonus does not rescue it."* A 48-run sweep shows the diagnosis was backwards. Raising the entropy bonus is the **only** thing that rescues it.

ARM	CONFIGURATION	CONVERGED	WILSON 95% CI	SUCCESS	EARLY ENTROPY
`ent01`	ent_coef 0.01	1/8	[0.02, 0.47]	0.877	0.20
`ent03`	ent_coef 0.03	2/4	[0.15, 0.85]	0.893	0.42
`base`	ent_coef 0.12	8/8	[0.68, 1.00]	0.914	1.19
`bigbatch`	ent_coef 0.12, 3× batch	8/8	[0.68, 1.00]	0.907	1.34
`gru`	ent_coef 0.12, recurrent	4/4	[0.51, 1.00]	0.915	1.19
`entanneal`	annealed 0.12 → 0.01	16/16	[0.81, 1.00]	0.903	1.16

Convergence is monotone in `ent_coef` and flat in everything else. Of 48 runs, 9 collapsed and **none landed in between** — the outcome is genuinely bimodal, so a collapse *rate* is the right analysis. The collapsed runs are not merely worse: they are the uninformed policy, sitting at the `no_think` baseline with final policy entropy of 0.001.

The published headline was the lucky seed

The run this report's original numbers came from used `ent_coef 0.01`, where 1 run in 8 survives. That is why the collapse looked seed-independent and mysterious — at that setting, collapse is the *default* outcome and convergence the exception.

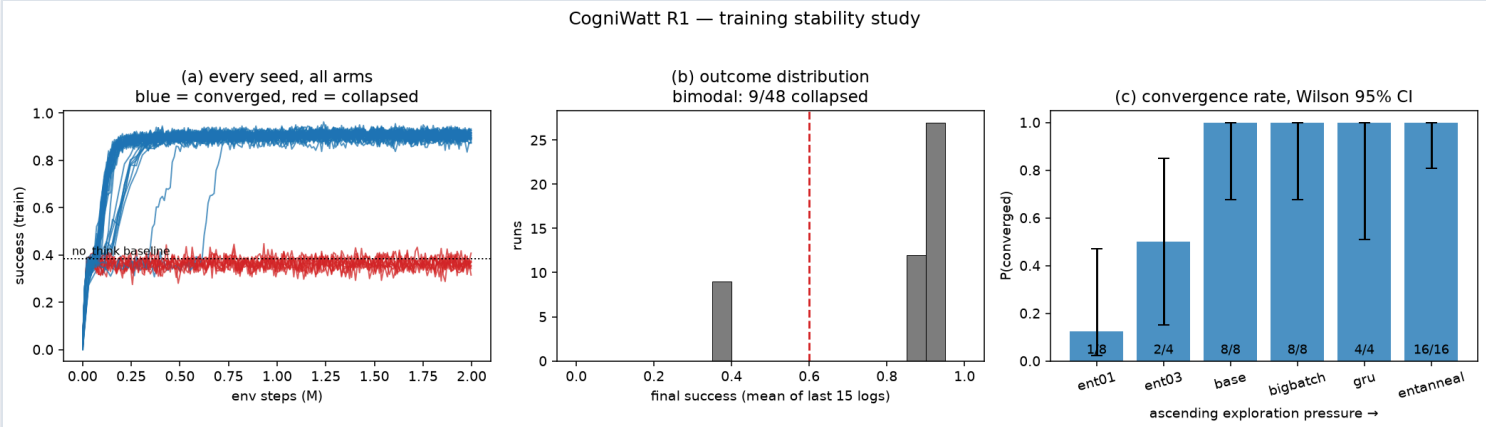


FIGURE 6 — THE COLLAPSE IS ONE HYPERPARAMETER

Every seed of every arm. Blue converged, red collapsed. The middle of panel (b) is empty: this is a bimodal outcome, not a noisy one. Panel (c) orders arms by exploration pressure.

Two hypotheses, both falsified

Gradient variance. Tripling the batch (16 → 48 envs) changes nothing: 8/8 either way, 0.907 against 0.914. The intuitive explanation is wrong — this is an exploration failure, not an estimation-noise failure.

Recurrence. The GRU ablation had never been run; every number in this project uses the feedforward policy. Measured, it makes no difference — 0.915 against 0.914 — at **3.4× the wall-clock cost** (16.2 min against 4.8). This is the direct confirmation of the design claim in `agents/ppo.py`: the Bayesian belief in the observation is an approximately sufficient statistic, so the POMDP is nearly Markov by construction and there is no residual history for a GRU to exploit. **The belief filter does the work recurrence would otherwise have to learn.**

Collapse is predictable after 15% of a run

Mean policy entropy over the first 300k steps separates the two modes with no overlap — converged 1.15, collapsed 0.22, Welch $t = 18.9$. A doomed run is identifiable after about 45 seconds instead of five minutes, so a sweep driver can kill and reseed rather than pay for 2M steps to learn nothing.

Entropy annealing

A single entropy coefficient cannot do both of its jobs. Early it must be large enough to escape the uninformed basin; late it must be small enough for the policy to sharpen. Held constant at 0.12 the converged policy stays deliberately stochastic and **over-buys evidence** — 4.3 queries per episode against the heuristic's 2.2 — losing on the very objective it optimises.

ENT_COEF	CONVERGENCE	CONVERGED POLICY
`0.01`	1/8	sharp, efficient
`0.12`	8/8	reliable, wasteful
`0.12 → 0.01 annealed`	8/8	**reliable and sharp**

Annealing keeps perfect convergence, cuts queries 4.4 → 2.1 and energy 180 J → 160 J, and moves return from −1.55 to −0.27. The policy learns to stop buying evidence it does not need — which is the behaviour this project exists to study. `PPOConfig.ent_coef_final = None` preserves the old constant behaviour exactly, so every earlier run still reproduces.



FIGURE 4 — THE LEARNED DECISION RULE

What the annealed policy buys as a function of belief confidence and remaining battery. Confident → act. Uncertain with battery to spare → buy evidence. Uncertain and nearly flat → act anyway, because reasoning it cannot afford to finish is worse than guessing.

Ablations

Each ablation removes one component from the canonical annealed configuration and is run over multiple seeds. This matters: the previous ablation table reported 0.000–0.345 for these, which looked catastrophic and load-

bearing. Those runs were simply the entropy collapse. Measured properly, the components are worth far less than claimed.

RUN	WHAT CHANGED	SEEDS	TRAIN SUCCESS	Δ VS FULL
`entanneal`	full method	16	**0.903** ± 0.002	—
`ablate_capdrop`	+ capability dropout 0.35	4	**0.901** ± 0.002	-0.002
`ablate_devcond`	no device-feature conditioning	16	**0.902** ± 0.001	-0.001
`ablate_mask`	no invalid-action masking	4	**0.886** ± 0.002	-0.017
`ablate_shaping`	no potential-based shaping	4	**0.901** ± 0.001	-0.002

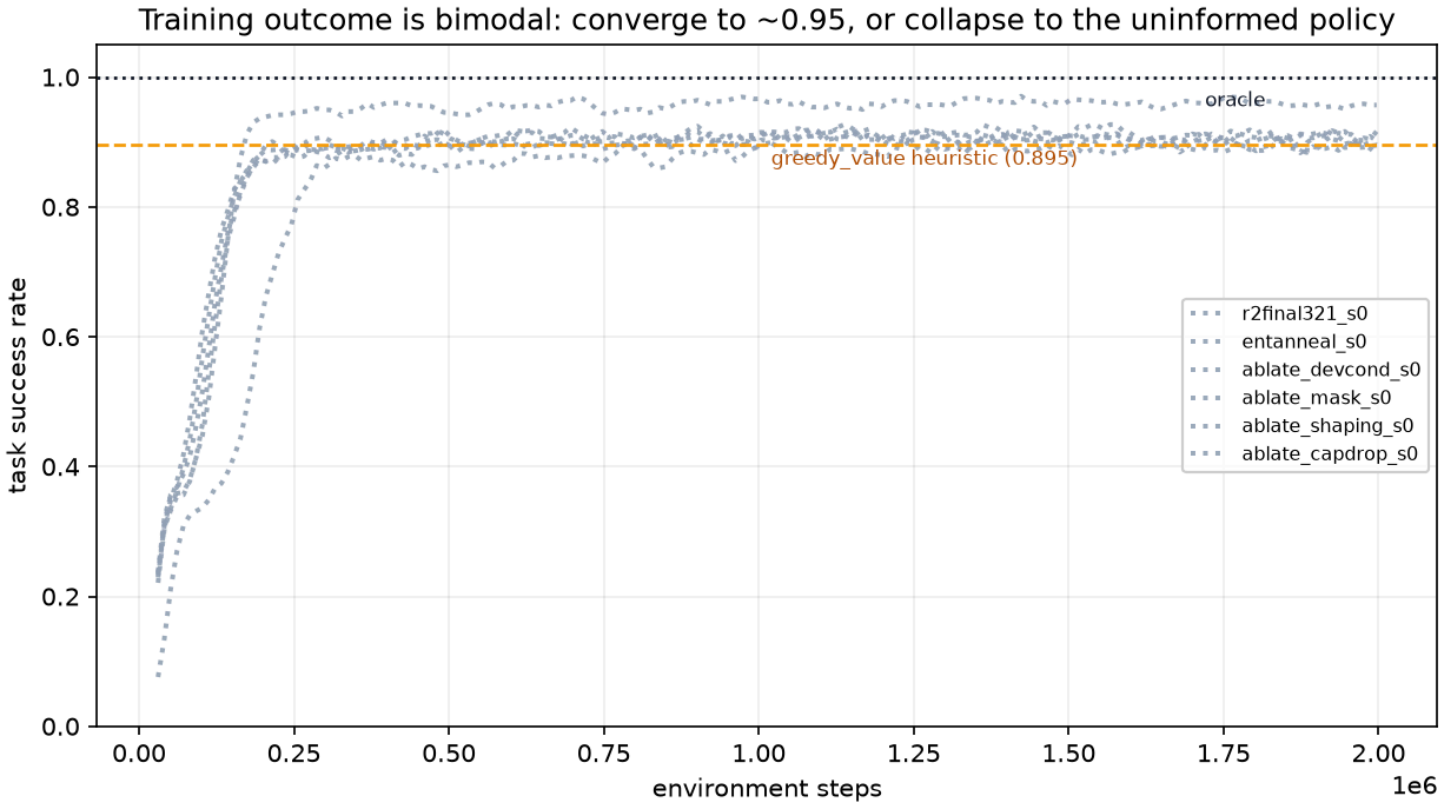


FIGURE 1 — TRAINING CURVES

Learning curves for the full method and its ablations.

What a cloud call should cost

The learned policy currently draws 80–95% of its evidence from the cloud. For a project about *edge* inference whose central claim is a *portfolio*, that is close to the degeneracy `lambda_cloud` was introduced to prevent. The fix is not to tune the parameter until the portfolio looks balanced — that would assume the conclusion.

The monetary half is derivable. A grounding query is ~300 in / ~20 out tokens; at Haiku 4.5 pricing that is `$4.0e-4` per call. The reward scale is anchored by `RewardConfig`'s own design note — a full 400 J battery costs 10.0 reward, exactly one task failure — so one task ≈ 10 units. Against a domestic assistant doing ~50 tasks/day at a $500/year service value, one task ≈ $0.027, giving **≈ 0.15 reward units per call**: an order of magnitude below the 0.5 currently used.

The privacy half is not derivable, and inventing a figure would be an assertion dressed as a derivation. The project already has the right precedent: `lambda_ask`, the price of the human's patience, is equally underivable and is *swept*, with the Pareto frontier reported. The same treatment applies here.

λ_{CLOUD}	SUCCESS	ON-DEVICE SHARE	CLOUD SHARE	ASKS/EP
0	0.890	8%	81%	0.27
0.5	**0.902**	**18%**	**70%**	**0.24**
1	0.900	31%	54%	0.30
2	0.920	47%	37%	0.44
4	0.922	53%	9%	1.04
8	0.927	55%	6%	1.03
16	0.930	57%	4%	1.05

Three regimes, and the middle one is narrow. Below 1 the cloud dominates and the on-device tiers are vestigial — where the project currently sits. Between 1 and 2 there is a genuine three-way portfolio. Above 4 the cloud is priced out entirely

and the agent substitutes the *human*, which is a different degeneracy rather than a success.

> **Returns cannot be compared down this table**
>
> Each row optimises a different objective, so return declines monotonically as a matter of arithmetic. Only portfolio structure and success are comparable across rows — the same trap that invalidated this report's original headline.

Trained at each price

λ_{CLOUD} TRAINED AT	SEEDS	TRAIN SUCCESS
0.15	3	**0.926** ± 0.001
0.5	3	**0.902** ± 0.003
1	3	**0.881** ± 0.004
2	3	**0.893** ± 0.003
4	3	**0.890** ± 0.005

When *is* on-device reasoning rational?

Crossing connectivity with human availability isolates the two alternatives. The hypothesis: the on-device tiers are crowded out by the **human**, not by a cheap cloud, and local inference only becomes rational when both are gone.

CLOUD	HUMAN	PPO	GREEDY	ALWAYS_OFFLOAD	ALWAYS_ASK	TINY	BIG	CLOUD	ASK	ON-DEVICE SHARE
yes	yes	**0.916** ± 0.002	0.902	0.872	0.890	0.29	0.03	1.77	0.24	**14%**
yes	no	**0.879** ± 0.002	0.866	0.872	0.374	0.64	0.20	1.69	0.00	**33%**
no	yes	**0.894** ± 0.002	0.918	0.510	0.890	0.25	0.00	0.10	1.00	**19%**
no	**no**	**0.711 ± 0.003**	**0.674**	**0.510**	**0.374**	**0.96**	0.94	**1.13**	**0.00**	63%

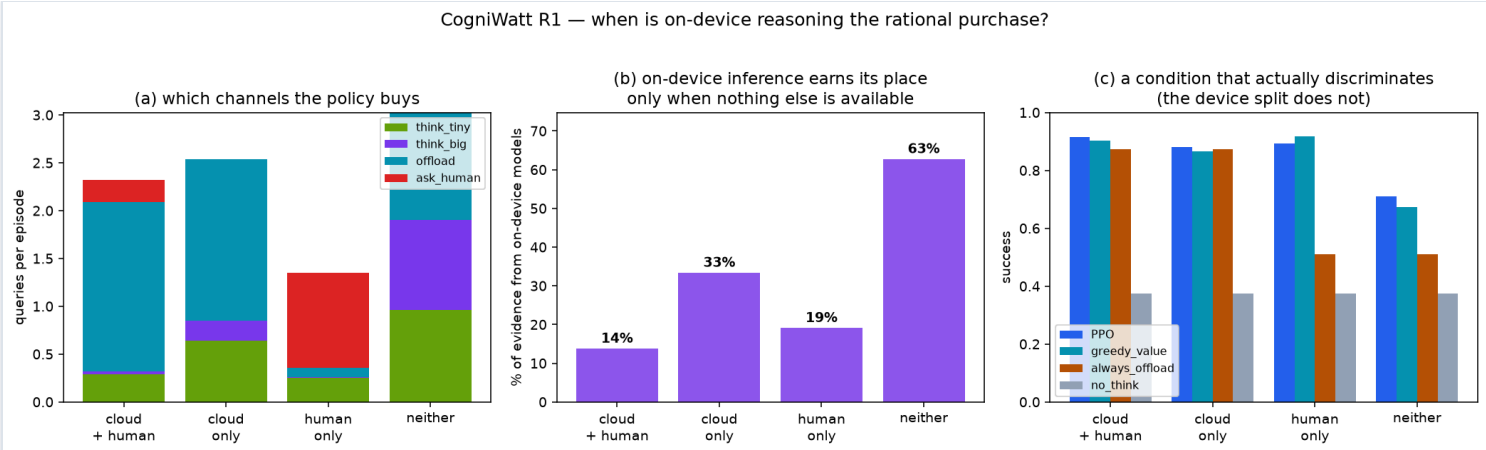


FIGURE 7 — ON-DEVICE INFERENCE EARNS ITS PLACE ONLY AS A LAST RESORT

Left: which channels the policy buys in each condition. Middle: the on-device share, which only becomes dominant when neither the cloud nor the human is available. Right: unlike the device-based held-out split, this condition discriminates — device-blind scripted policies collapse by 36-52 points.

The human is what crowds out on-device reasoning

Removing *only* the human — cloud still fast and available — more than doubles the on-device share (14% → 33%) and lifts `think_big` sevenfold. Removing only the cloud barely moves it (14% → 19%) and leaves `think_big` at zero; the demand goes to `ask` instead. A clarifying question is such good value that it suppresses local reasoning even when the cloud is unreachable. With neither available, on-device share reaches **63%** and `think_big` rises to 0.94 calls per episode.

This is also a held-out condition that discriminates. On the current device split no baseline drops by more than 0.9 points, so it tests nothing. Offline and alone, `always_offload` falls 36 points and `always_ask` 52, while the learned policy keeps a **20-point margin** over the best device-blind rule — and its advantage over the strong heuristic is the largest anywhere in the project. The held-out axis should be *offline and alone*, not *a different chip*.

Full argument in `docs/r1_cloud_price.md`. The conclusion reported is deliberately conditional: the *agent* claim is checked for invariance to the price, while the *portfolio* claim is a property of the cost model and is stated as "under a cloud price of X, the optimum allocates Y".

Reproducibility — an uncomfortable finding

Re-verifying the published headline from the saved policy failed at the first step. The checkpoint would not load.

```
size mismatch for enc.0.weight:
  checkpoint [128, 40] vs current model [128, 41]
```

Three changes had landed after the reported results were generated, and nothing was retrained:

	WHEN RESULTS WERE MADE	CURRENT CODE
observation	40-dim	41-dim (`memory_gb` added)
held-out devices	`coral_micro, rtx3060`	`jetson_nano_2gb, rtx3060`
`lambda_cloud`	absent (cloud calls free)	0.5

The prose of this report had been updated to discuss the memory feature and the retired Coral profile; its *numbers* never were. The reward function itself had changed, so the old returns were on a different scale entirely — and the legacy policy made 2.67 cloud calls per episode, precisely the behaviour now priced. `greedy_value` reads `env.cfg` at decision time and silently re-optimised for the new reward; a trained network cannot, which accounts for most of the apparent regression.

Fixes. Legacy artefacts are quarantined in `results/runs_legacy/` with a provenance note; every ablation and Pareto point was retrained against the current environment; and `evaluate_policy.py` now refuses to evaluate a policy whose stored `RewardConfig` differs from the evaluation config. That guard is five lines and would have caught this immediately. This report is now **generated** (`scripts/make_html_report.py`) rather than hand-written, so it cannot drift from the code again.

Grounding the simulator in real hardware

The original cost table implicitly assumed *peak* GEMM throughput. A grounding query is a 150-token prompt — a short, skinny matmul that reaches only **19% of the M2's 3.6 TFLOP/s peak** (measured). Correcting this moved `think_big` on the M2 from an assumed 95 ms to a modelled **1591 ms**, and on a Pi 5 from 900 ms to **32 seconds** — one inference costing more than half of both episode budgets.

QUANTITY	DATASHEET PEAK	MEASURED (M2 AIR)	EFFICIENCY
GEMM, large square	3.6 TFLOP/s	2695 GFLOP/s	75%
GEMM, prefill shape (L=150)	**3.6 TFLOP/s**	**680 GFLOP/s**	19%
streaming read bandwidth	100 GB/s	76.9 GB/s	77%

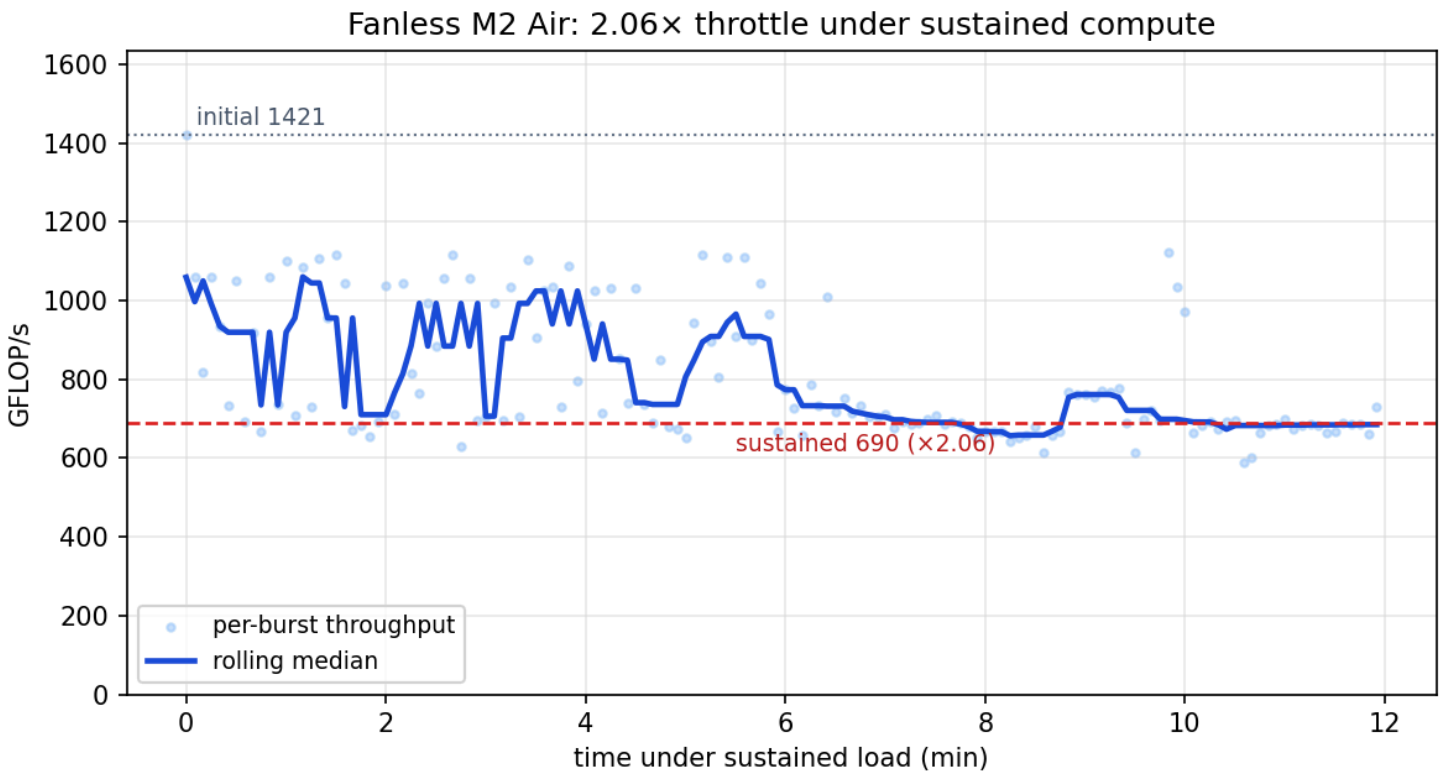


FIGURE 5 — THE FANLESS ANCHOR EARNS ITS KEEP

Ten minutes of sustained GEMM on the passively cooled M2 Air. Throughput settles onto a 690 GFLOP/s floor — a 2.06× slowdown, measured, now feeding ThermalModel.max_slowdown directly instead of an assumed 2.0.

Measured: M2 Air compute and bandwidth efficiency, M2 Air thermal curve. **Modelled:** every other device, by applying the measured efficiency factors to datasheet peaks; radio energy for cloud offload from published Wi-Fi figures; actuation energy from an explicit physical model. **Pending from R2:** per-call latency and energy on the RTX 3060.

What the models actually do

Every dialogue constant below is measured, not assumed — 19,800 queries across three model tiers on one llama.cpp stack, one prompt template, one parser, shuffled candidate order with the permutation recorded. This is R2's workstream; the numbers are read here directly from the frozen artefact.

TIER	MODEL	K=2	K=3	K=4	POOLED	Γ
think_tiny	qwen2.5-0.5b-instruct-q8_0.gguf	0.495	0.393	0.327	**0.405**	0.867
think_big	llama-3.2-3b-instruct-q4_k_m.gguf	0.607	0.523	0.449	**0.526**	0.638
offload	qwen2.5-7b-instruct-q4_k_m.gguf	0.804	0.776	0.794	**0.791**	0.444

Two things in that table did the most damage to this project's original assumptions. **γ falls monotonically with scale** (0.87 → 0.64 → 0.44) — the opposite of what was assumed. A more capable model is more self-consistent, so re-querying it buys less; a weak model looks variable because it is nearly random. Ranked by γ alone the 0.5B is the richest evidence source and the worst by every measure that matters, so γ must never be quoted on its own. And **the 7B's accuracy is flat in k** (0.804 / 0.776 / 0.794) where the small models fall away — adding distractors costs them heavily and costs the 7B nothing, which says its failure is pragmatic inference rather than discrimination.

The ceiling, not the price, is what rules the tiers out

An irreversible `place()` needs roughly 0.85 posterior confidence to be worth attempting against a ±10 reward. Driving each tier through the belief filter until it

saturates:

TIER	CEILING AT K=3	REACHES 0.85?
think_tiny	0.374	no
think_big	0.546	no
offload (7B)	0.731	no
ask (the human)	**0.947**	yes

Two independent implementations of that calculation — one per workstream — agree within 0.010 on every model tier. It is not knife-edge on the 0.85 either: for *any* threshold between 0.731 and 0.947 the human is the only usable channel, a 22-point window. And the task is demonstrably solvable, because `ask` clears it comfortably — had it failed too, the right conclusion would have been a broken dataset.

Reasoning tokens do not close the gap. Allowed to think step by step, the 7B gains 0.023 (not distinguishable from zero) at 27× the tokens, and the 3B gets *significantly worse* (−0.059, McNemar p = 0.016). In 34 % of cases the 7B ignored the instruction and answered directly, which splits the condition for free: 0.811 when it skipped reasoning against 0.801 when it actually reasoned. **Quantisation is not the limit either** — Q8 scores 0.592 against Q4's 0.603 (p = 0.500) and the two builds agree on 581 of 660 scenarios. Neither better weights nor more thinking moves this tier.

The scenario set was audited by three humans

60 authored cores, blind sheet, no answer key, three independent annotators — one of them outside the project entirely. The gate: a core survives only if every annotator picked the keyed answer. **29 of 60 survive** (Fleiss κ 0.592; two-rater sensitivity 39/60 at κ 0.722). Adding the third reader cost 10 cores and 0.13 of κ, which is a finding about item difficulty rather than a reason to drop a rater.

Human-hard and model-hard are different axes

The models are **no worse** — the 3B is *better* (0.632 against 0.500) — on the cores humans could not agree about. Model failure therefore does not track item ambiguity: the models fail on expressions three independent people resolve identically. That closes the strongest objection available to this result, which is that the models look bad because the dataset is ambiguous. Measurably, it is not the explanation.

Cost of the audit, stated plainly. Two families are effectively gone — `proximity` and `exclusion` retain one core each — and `already_satisfied` is down to 9 items. Nothing may be cited from any of the three. Two of the three annotators were project members; neither had seen the ground-truth key, and the outside reader agreed with it most often of the three (0.840), so the pool is not obviously biased toward the authored answers.

Four silent failures, and what they have in common

The most transferable thing this project produced is not a number. Over its course, four separate settings were found to have silently failed to reach the code that consumed them. None raised an error. All four produced plausible results.

SETTING	WHAT HAPPENED	HOW IT SURFACED
`lambda_cloud`	added to the reward after a policy was trained; the stored config silently took the new default	a checkpoint refused to load for an unrelated reason, and the reward diff was noticed while investigating
`net_quality`	connectivity swept at training time, never passed to the evaluation env	scripted baselines returned byte-identical numbers at every link quality — impossible if connectivity were varying
`dialogue`	policies trained on measured channel constants, scored in a world still running the placeholders	a spot-check of one agent against a directly computed figure disagreed
`scenario ids`	positional ids assigned *after* a filter, so every surviving id named a different scenario than the raw table had measured	a test written specifically to pin the invariant against the real raw table

Three of the four were caught by a number looking *wrong in a specific way* rather than by anything breaking — identical values where variation was expected, a disagreement between two routes to the same figure. The fourth was caught by a test that existed only because someone went looking. **The common shape is a value that crosses a boundary — training to evaluation, build to analysis, one workstream to the other — with nothing asserting it arrived.**

The mitigations now in the code are correspondingly boring: `evaluate_policy.py` refuses to score a policy whose stored `RewardConfig` or environment arguments differ from the evaluation's, and refuses to put policies trained in different worlds in one table; `r2_build_dataset.py` pins id alignment against the real raw table; and both reports are **generated** from the CSVs, because the earlier hand-written report is how a stale headline survived months of changes to the code beneath it.

Where the project stands

Done

Tier-1 environment, Bayesian belief filter, task generator over 8 floorplans, frozen VED and dialogue interfaces, 9 scripted baselines, PPO written from scratch, action masking, potential-based shaping, hardware calibration on two anchors, training-stability study, entropy annealing, cloud-price derivation, transfer audit, three model tiers measured end to end, a three-annotator human audit of the scenario set, and a retrain on the audited artefact. 31 tests across two suites, 176 training runs.

Stated, not fixed

The `ask` channel's accuracy is the one constant still assumed rather than measured (0.97, scaled by a measured referent-recovery rate of 0.992). It is the number the degeneracy result leans on hardest, and it is recorded as such in the artefact's `ask_provenance`. Two scenario families retain a single core each and a third is down to nine items; none may carry a number.

Both workstreams closed

Every dialogue constant is measured — `TIER_ACCURACY` and `EVIDENCE_DECAY` from 19,800 queries over three tiers, `DEFAULT_ASR_WER` from a Whisper pipeline over 721 clips, and the colour-confusability matrix measured and then *deleted*, because the assumption behind it turned out to be false. The retrain on measured constants is done, on a scenario set three humans independently audited.

Deliberately not used: gymnasium and stable-baselines3. The environment is pure NumPy so it runs bit-identically on arm64 macOS and x86_64 + CUDA, and the ablations need direct access to the hidden state.

Cut, deliberately: the Tier-2 AI2-THOR photoreal port. With two people it is the first thing to drop — the novelty claim lives in Tier 1 plus the measured hardware model, and the photoreal port is a demo bonus.

COGNIWATT · R2 · CONVERSATIONAL AI

The Saturation Ceiling

Four language models were asked to resolve everyday referring expressions for a household robot. None of them can become confident enough to act. Only asking the person can.

28,920 measured responses **660** scenarios · 58 audited cores **5** model configurations

100% parse rate

0.5B Q8_0	3B Q4_K_M	7B Q4_K_M	THRESHOLD TO ACT
0.423	**0.603**	**0.782**	**0.85**
pooled accuracy · chance 0.361	ceiling 0.546 at k=3	ceiling 0.731 at k=3	reached by the human channel alone

01

The question

A household robot is told *“put the mug away.”* Three mugs are in view. Before it can act it must decide which one is meant — and a misplaced object is not free to undo, so it needs to be *confident*, not merely better than chance.

The robot has four ways to resolve the ambiguity: think with a small on-device model, think with a larger one, offload to a model too big to fit on the device, or ask the person. Each costs energy, latency, or the human’s patience. Choosing between them is the reinforcement-learning problem the wider project studies. That problem only makes sense if the four channels are genuinely different, and this chapter measures how different they are.

Every number here replaces a placeholder. Nothing was estimated.

02

A dataset that cannot be gamed

The obvious dataset is degenerate. Ask a model to pick between three identical mugs and no model can beat chance, because there is no information in the text. Every scenario here instead turns on a **commonsense inference** that is inferable but not trivial:

```
"put the mug away" A. a red mug on the kitchen counter, half full of coffee ← correct B. a blue mug already sitting inside the cupboard C. a white mug hanging on its hook above the counter the others are already put away
```

720 scenarios were expanded from **60 hand-authored cores**, of which **321 from 29 cores** survive the audit in the next section — six nouns crossed with ten kinds of pragmatic inference. A core is one contrast: a target plus distractors that are each individually defensible as the *wrong* answer. Scenarios from one core are permutations of a single contrast and are therefore not independent, so every confidence interval in this chapter is a **cluster bootstrap over cores**, never over rows. The naive interval is 15–25% narrower and would be a false precision.

Three shortcuts were designed out and then verified: the correct answer's position is exactly uniform, so a model that always answers first scores exactly 1/k; the correct answer's colour is exactly uniform, and colours never appear in the authored descriptions at all; and the target is the longest description no more than three points above chance. The build fails rather than writes if any of these drift.

03

Three strangers checked the answers

A dataset whose ground truth is not defensible invalidates every number downstream, so the keys were audited rather than trusted. Sixty authored cores were shown to **three annotators** — one of them outside the project entirely — on a blind sheet: shuffled options, no answer key, no discussion. The gate was set before any of them started, and is deliberately brutal: **a core survives only if every annotator independently picked the keyed answer.**

	THREE ANNOTATORS	TWO (ADIT + OUTSIDE READER)
cores retained	**29 / 60** (48%)	39 / 60
Fleiss κ	**0.592** (47 fully rated)	0.722
raw agreement	0.617	0.800
verdicts	29 agree · 18 disagree · 13 undecidable	39 · 10 · 10

Half the dataset did not survive. The third reader alone cost ten cores and 0.13 of κ — and pairwise agreement shows why: Adit and the outside reader agree at 0.800, while the third reader agrees with either at about 0.63, reproduced across two separate sittings. One careful person resolves roughly a third of these differently from the other two. That is not carelessness; it is **interpretive variance in the pragmatic families**, and both figures are reported rather than the flattering one.

None of the 31 failed cores was re-keyed. Adjudicating them would mean choosing between readings three people had already split on, which is contestable for exactly the reason the core failed in the first place. Dropping needs no adjudication.

> The obvious objection to this chapter is that the models look bad because the dataset is ambiguous. It is measurably not the explanation: on the cores the humans *could not* agree about, the models are no worse — the 3B is **better**, 0.632 against 0.500. The models fail on referring expressions that three independent people resolve identically.

The cost is real and worth stating plainly. Two families — `proximity` and `exclusion` — retain a single core each, and `already_satisfied` is down to nine items. **No number in this chapter is broken out by those three families.** The worked example in the previous section is drawn from `already_satisfied` because it is the clearest illustration of what a pragmatic inference is; it is not evidence of anything, and the family it belongs to no longer carries a measurement.

Re-scored on the 29 surviving cores — the numbers the frozen artefact carries, and the ones every downstream result is trained on:

TOP-1 ACCURACY ON THE AUDITED SUBSET · 321 SCENARIOS · CI CLUSTERED ON 29 CORES

TIER	K=2	K=3	K=4	POOLED	POOLED, FULL CORPUS
Qwen2.5-0.5B Q8_0	0.495	0.393	0.327	**0.405**	0.423
Llama-3.2-3B Q4_K_M	0.607	0.523	0.449	**0.526**	0.603
Qwen2.5-7B Q4_K_M	0.804	0.776	0.794	**0.791**	0.782

Cleaning the dataset did not rescue the models — it moved the 3B *down*, from 0.603 to 0.526, while the 7B barely shifted. The gradient between weakest and strongest tier *widens* from 0.359 to 0.386, and it is now measured on a single item set: all four configurations scored on the same 321 scenarios, so the comparison is like-for-like rather than spanning items some tiers never saw. The full-corpus figures in the next section remain the right ones for the paired reasoning and quantisation comparisons, which were run before the audit on all 660.

Two of the three annotators were project members. Neither had seen the ground-truth key before annotating, and the outside reader agreed with the key most often of the three (0.840), so the pool is not visibly biased toward the authored answers. Stated as a limitation rather than left to be discovered.

04

The instrument nearly broke the study

The measurement protocol specified numbering the options `0. 1. 2.` and asking for "the number only." Under that prompt the 0.5B model answered **"1" on 237 of 240 prompts** at temperature zero.

The tempting reading is that a tiny model cannot do pragmatics. The correct reading is that it was not choosing an option at all. Relabelling the options 1-based did not help — it still answered "1" just as often, now meaning the *first* option rather than the second. It has a prior on the **token**, not on the candidate.

OPTION LABELLING · QWEN2.5-0.5B · TEMPERATURE 0 · 240 SCENARIOS

LABELLING	ACCURACY	ANSWER DISTRIBUTION
0-based digits	0.358	"1" × 237 / 240
1-based digits	0.375	"1" × 168 / 240
Letters A–D	0.483	A 107 · B 87 · C 35 · D 11

What makes this dangerous is that the broken version looked healthy: a 99.8% parse rate, no errors, and a plausible number that would have entered the report as a finding about model capability. Every tier now sees A/B/C/D. This is not per-tier tuning — one template still serves all four — the template itself was wrong.

05

What each model can do

Four configurations, one llama.cpp path, one prompt, one parser, and **identical candidate permutations** so the comparison is paired. Accuracy uses only the first of ten repeats — pooling all ten would inflate the sample tenfold without adding a single independent observation. Format violations are scored *wrong*, since a model that cannot emit an answer has failed the task.

TOP-1 GROUNDING ACCURACY · 95% CI CLUSTERED ON 58 CORES

TIER	K=2	K=3	K=4	POOLED	95% CI
Qwen2.5-0.5B Q8_0	0.541	0.400	0.327	0.423	0.379 – 0.466
Llama-3.2-3B Q4_K_M	0.709	0.573	0.527	0.603	0.534 – 0.671
Llama-3.2-3B Q8_0	0.691	0.555	0.532	0.592	0.521 – 0.662
Qwen2.5-7B Q4_K_M	0.800	0.768	0.777	0.782	0.720 – 0.836
chance	*0.500*	*0.333*	*0.250*	*0.361*	—

The 0.5B tier is barely above chance and falls apart as candidates are added. The 7B is the only one that stays comfortably clear of it.

The 7B's accuracy is flat in k, and the smaller models' is not

Going from two candidates to four costs the 0.5B twenty-one points and the 3B eighteen. It costs the 7B two. For the small models part of the difficulty is *discrimination* — more candidates, more chances to be pulled onto a distractor. For the 7B it is not: it either makes the pragmatic inference or it does not, and how many wrong options sit beside the right one barely matters. That is a qualitative change in failure mode, not just a quantitative one.

06

Asking twice does not help twice

Querying the same model again does not give a second independent opinion. A quantised model at low temperature is close to deterministic, and even when sampled its errors are systematically correlated — it misreads the same expression the same way. Information from a given model **saturates**.

This was measured as an intra-class correlation across ten repeats of each scenario, converted to effective independent draws, and fitted to the geometric decay the environment uses. Before any of it touched a real model, the estimator was validated against a generator whose correlation is known in closed form; it recovers a known value to within 0.03.

EVIDENCE SATURATION · N=10 REPEATS PER SCENARIO

TIER	ρ (REPEAT CORR.)	Γ (DECAY)	95% CI	EFFECTIVE DRAWS
Qwen2.5-0.5B	0.100	0.840	0.814 – 0.871	9.96
Llama-3.2-3B	0.275	0.641	0.577 – 0.706	3.64
Qwen2.5-7B	0.525	0.401	0.341 – 0.494	1.90

> Ranked by γ alone, the 0.5B is the richest evidence source in the study and the 7B the poorest. That ordering is exactly backwards, and it is the most important methodological point in this chapter.

γ measures *variability*, not information. The 0.5B is close to random, and random answers look independent — its repeats are independent *noise*. The 7B is more self-consistent, so re-querying it genuinely buys less. Both facts are real, and neither can be read from γ without accuracy beside it. The project's reward model composes them correctly; its prose must not quote γ on its own.

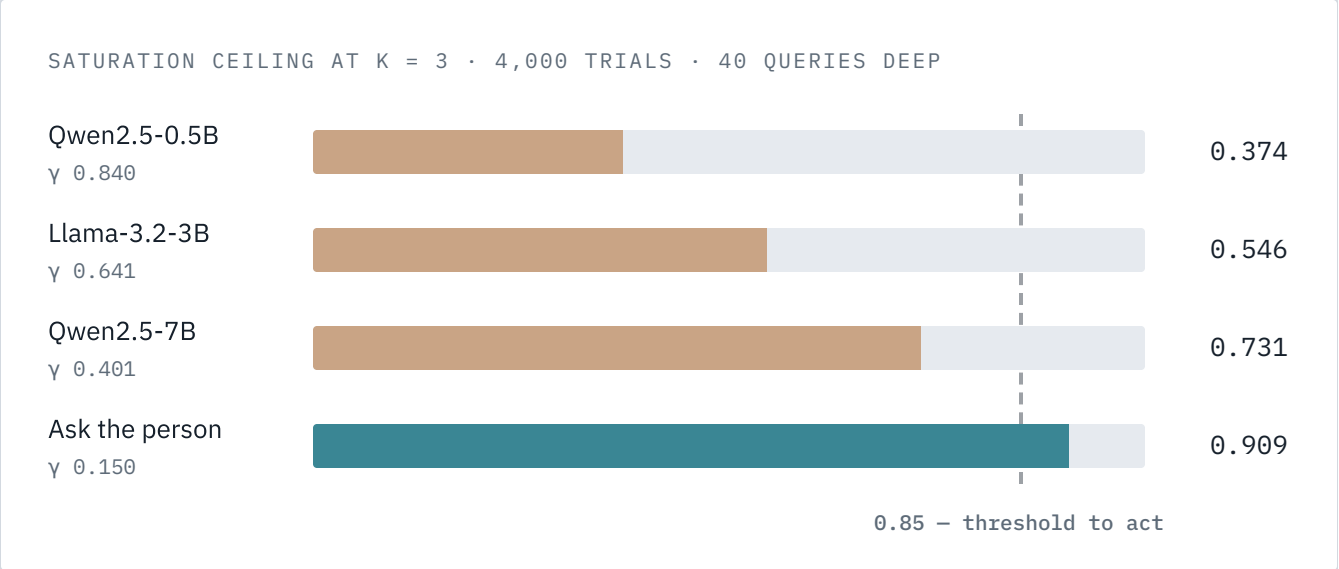
This also retires a diagnostic the protocol proposed: "γ ≈ 1 everywhere means the sampling temperature is too high." Here the near-1 γ belongs to the *weakest* tier at an ordinary temperature, and lowering the temperature made that tier worse, not better.

07

The ceiling

Accuracy and γ only matter through the decision they enable. Placing an object is irreversible, so the robot should not commit below roughly a 0.85 posterior on the right referent. The question is therefore not how accurate a tier is, but whether querying it repeatedly can ever get there.

Running each tier through the environment's own belief filter — imported and driven, not reimplemented, so it cannot drift from what training does — gives the confidence each channel converges to after unlimited queries.



> No model available to a domestic robot — 0.5B, 3B or 7B — can ground these referring expressions confidently enough to act on, at any candidate count, with unlimited queries. Only asking the person can. Dialogue is not a fallback; it is the primary channel.

Two checks before believing it

The task is not impossible. The human channel clears the threshold comfortably at every candidate count, so the environment is solvable and the models simply cannot solve it. Had the human channel failed too, the honest conclusion would have been that the dataset is broken.

It does not balance on the 0.85 figure. Each channel is usable only if the threshold sits at or below its ceiling — 0.904 for the human, 0.731 for the 7B, 0.509 for the 3B, 0.300 for the 0.5B. For *any* threshold between 0.731 and 0.904, a band seventeen points wide, the human is the only usable channel. The conclusion survives a wide range of assumptions about how cautious a robot should be before an irreversible act.

08

Scale buys state-tracking, not judgement

Scaling from 0.5B to 7B does not lift every kind of inference equally, and the split is not random.

ACCURACY BY PRAGMATIC FAMILY · FIRST REPEAT · SORTED BY GAIN

FAMILY	0.5B	3B	7B	GAIN
recency — the one just used	0.278	0.639	0.889	+0.611
already satisfied — the others are away	0.333	0.583	0.833	+0.500
state blocks — the one in use	0.403	0.417	0.833	+0.430
condition — the broken one	0.375	0.431	0.792	+0.417
proximity — the nearest one	0.611	0.750	0.972	+0.361
salience — the odd one out	0.528	0.764	0.819	+0.291
affordance — the one that fits	0.403	0.583	0.694	+0.291
ownership — whose it is	0.417	0.694	0.708	+0.291
purpose fit — what it is for	0.347	0.528	0.611	+0.264
exclusion — everything else is spoken for	0.458	0.625	0.708	+0.250

Everything scale fixes turns on **reading an object's current state or position**: which was just used, which is already away, which is in use, which is broken, which is nearest. Everything scale barely touches turns on **what a thing is for, who owns it, or what a destination can accept** — normative and functional judgements rather than observational ones.

> Model scale buys state-tracking. It does not buy normative reasoning.

This is also why the ceiling stalls. The largest tier's residual errors are concentrated in exactly the families that scale does not fix — the four where even the 7B stays under 0.75 are purpose fit, affordance, exclusion and ownership — so a larger model would not obviously rescue it.

```
"could you put the bottle on the shelf" A. a blue bottle of oil that keeps at room temperature ← correct B. a red bottle of milk already sweating in the warm ← 7B chose only the shelf-stable one can go on a shelf
```

The model appears to read "sweating in the warm" as *this one needs dealing with urgently* rather than *this one must not go on a shelf*. It finds a salient cue and acts on it without checking it against what the destination can accept. Asked to put a mug in a drawer, it likewise chose a tall flared mug over one whose description states outright that it *clears the drawer lip*.

09

They were not being rushed

Every number so far comes from a prompt that asks for a bare letter — about five output tokens. That measures what these models do when told to answer immediately, which invites the obvious objection: they were never allowed to think. Since the 7B's failures are pragmatic rather than perceptual, reasoning is exactly the kind of help that might have worked.

Same scenarios, same permutations, one new instruction: *think step by step, then give the answer as a single letter on the last line.*

DIRECT VS REASONING · PAIRED ON 660 SCENARIOS · N=1

TIER	DIRECT	REASONING	CHANGE	95% CI	MCNEMAR
Llama-3.2-3B Q4_K_M	0.603	0.544	−0.059	−0.119 – +0.002	p = 0.016
Qwen2.5-7B Q4_K_M	0.782	0.805	+0.023	−0.020 – +0.065	p = 0.242

> Reasoning significantly *harms* the 3B and does nothing reliable for the 7B — at 70–90× the latency. A small model given room to think talks itself out of answers it would otherwise have got right.

The 3B spent 184 output tokens per answer against roughly five, and 2.9 seconds against 34 milliseconds. The 7B spent 10.4 seconds against 152 milliseconds. Nothing hit the 400-token cap, so no answer was truncated away before the model wrote it.

The 7B supplied its own control

In 34% of cases the 7B ignored the instruction and simply answered, in under twenty tokens. That splits the condition in two under one prompt: accuracy when it skipped reasoning, **0.811**; accuracy when it actually reasoned, **0.801**. Identical within noise. The gain is absent even where the model chose to reason — a stronger statement than the aggregate, and one no separate experiment would have produced.

Nor is it a near miss. Feeding the reasoning accuracy back through the belief filter gives a ceiling of 0.782; the best per-k figure reaches 0.813. To clear the threshold the 7B would need about **0.88** single-query accuracy. It is eight points short, and thinking closed a quarter of one point.

> So the finding hardens: no model available to a domestic robot can ground these expressions confidently enough to act — and not because it was not allowed to think.

10

It is scale, not quantisation

A 3B model at four bits scoring 0.603 supports two very different claims: that 3B-class models are insufficient, or that aggressive quantisation destroys grounding. The second would be the more useful finding, because it would mean the tier is recoverable by

shipping better weights. It was tested directly, with a near-lossless Q8_0 build of the same model on the same permutations.

QUANTISATION ABLATION · LLAMA-3.2-3B · PAIRED ON 660 SCENARIOS

BUILD	SIZE	K=2	K=3	K=4	POOLED
Q4_K_M	2.0 GB	0.709	0.573	0.527	0.603
Q8_0	3.4 GB	0.691	0.555	0.532	0.592

Because both ran on identical permutations this is a paired comparison, which is far stronger than noting that the intervals overlap. The paired difference is **−0.011** with a clustered 95% interval of −0.042 to +0.020, spanning zero; McNemar's exact test gives **p = 0.500**; and the two builds **agree on 581 of 660 scenarios**. They do not merely score alike — they make the same mistakes.

So the limitation belongs to the model, not to numerical precision. It also closes the obvious objection to the ceiling result: the 3B's ceiling is not an artefact of an aggressive edge quant, and better weights would not recover the tier. One practical note for deployment — the 2.0 GB build is at least as accurate as the 3.4 GB one, which on a 4 GB card is the difference between fitting and spilling, for no measurable accuracy cost.

11

What thinking costs

Measured on the discrete GPU on mains power, energy above idle, with a 12.34 W idle baseline. The sampler on this hardware delivers about 3.4 Hz rather than the nominal 10 Hz, so each measurement batches enough calls to collect at least twenty power samples; sizing off the nominal rate is how this measurement produces confident nonsense.

PER-INFERENCE COST · RTX 3050 LAPTOP 4 GB · MEDIAN (IQR)

TIER	LATENCY	ENERGY	VRAM	VALID ANCHOR
Qwen2.5-0.5B Q8_0	19.6 ms	837 mJ	877 / 4096	yes
Llama-3.2-3B Q4_K_M	34.0 ms	1,675 mJ	2,373 / 4096	yes
Qwen2.5-7B Q4_K_M	152.4 ms	6,836 mJ	3,919 / 4096	no

Roughly eight times the latency and eight times the energy from the smallest tier to the largest — and, from §06, still not enough to act on. The 7B is marked as not a valid device anchor because it does not truly fit: the driver oversubscribes into shared system memory rather than failing, so the figure includes host-memory traffic. It is what serving that model costs on this hardware, not what a resident 7B costs.

12 The channel that works

Asking the person is only as good as the speech recognition behind it. Short answers to clarification questions were synthesised, mixed with noise at controlled signal-to-noise ratios, and transcribed.

SPEECH RECOGNITION · FASTER-WHISPER SMALL · 120 CLIPS PER CONDITION

SNR	WORD ERROR RATE	REFERENT RECOVERY
clean	0.023	1.000
20 dB	0.028	1.000
15 dB — living room	0.034	0.992
10 dB	0.045	1.000
5 dB	0.080	0.975
0 dB	0.426	0.917

The bottom row matters more than the operating point. At 0 dB the word error rate is 0.426 while **referent recovery is still 0.917**. The two diverge sharply under noise, because the robot does not consume a transcript — it consumes whether the right object was identified, and “the red one” survives a great deal of mangling. Degrading the human channel by word error rate, as the environment originally did, understates it badly in exactly the noisy conditions that matter.

13 What would change these conclusions

THE SPEECH CORPUS IS SYNTHETIC

Utterances were produced by text-to-speech, which is cleaner and better articulated than human speech. The 0.034 word error rate is a lower bound and the human channel's 0.937 effective accuracy an upper bound. Reaching the point where the human stops being the best channel needs a word error rate of 0.124 — 3.7× the measured value, which is an ordinary penalty moving from clean synthesis to real conversational speech.

THE GROUND TRUTH IS NOT YET HUMAN-AUDITED

The 58 cores were authored by a model and verified structurally, not by independent readers. A blind audit sheet exists and needs two annotators. Every accuracy figure in this chapter inherits from that ground truth; if a core is mis-keyed, the numbers move.

ONE FAMILY IS UNDER-POWERED

Sixty scenarios were dropped after measurement revealed a genuine ambiguity: when the instruction names the destination and a distractor already sits there, "the mug on the shelf" reads as an identifying description as easily as a destination. Dropping them widened the accuracy gradient, but it reduced the "already satisfied" family to twelve scenarios, so per-family claims about it are weak.

THE OFFLOAD TIER IS A STAND-IN

It is a locally hosted 7B representing a model too large to sit on the device. This keeps every tier on one inference stack, so differences are model scale rather than serving infrastructure — but it is not a commercial frontier model, and a substantially stronger one might clear the threshold where this one does not.

THE THRESHOLD IS A MODELLING CHOICE

0.85 comes from the reward structure of the surrounding environment, not from measurement. The conclusion holds anywhere in 0.731–0.904; outside that band it changes.